



Department of Computer Science and Business Systems
Academic Year 2023 – 2024(Odd Semester)

Degree, Semester & Branch: B.Tech., I & CSBS

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member: Ms.K.Usharani AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit 3 / linear search and binary search
- **Course Outcome:** CO 3 – Apply control structures, functions and string to write simple program for solving problems.
- **Topic Learning Outcome:** TLO10 - Students will be able to write the simple program using conditional, looping, string and functions concepts.
- **Activity Chosen:** Think Pair Share
- **Justification:**
 1. Think Pair Share is an active learning and also a collaborative learning method which increase the students focus and involvement on learning the topic.
 2. Sharing their knowledge with their teams will improve their understanding in the topic and also their communication skill.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
 1. **Material used :**
 - a. Textbook - Reema Thareja, “Python Programming using problem solving approach”, Second Edition, Oxford University Press, 2019
 - b. IDLE – to execute the program.
 2. **Prerequisite :** Knowledge on the simple programming constructs and searching.
 3. Activity comprises of 3 parts – Think, Pair, Share
 4. **Think** – The students are asked to think logic and code for linear search individually.
 5. **Pair** – Students are combined as group of 2 members to discuss their knowledge and also to get clarity on the topic.
 6. **Share** – Based on the understanding on the topic of students, some of the teams were chosen to share their knowledge on the topic to their class students.
 7. Finally the faculty consolidated the discussions on the topic .

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO3
CO5	2	1	1	3	3	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PO12	PSO3
	2	1	1	3	3	2	3
Justification for correlation	Apply the knowledge of engineering fundamentals like searching	Identify and analyze complex engineering problems related to searching.	Design solutions for complex engineering problems	Function effectively as an individual and as a member in a team	Communicate effectively on complex engineering activities with engineering community.	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning.	Apply appreciable knowledge of management skills cultured through real time practice.

- **Images / Screenshot of the practice:**





- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students felt that they can able to understand and get clarity on the topic very easily by sharing their knowledge with their friends.

- ❖ ***Benefit of the practice:***

Students were actively participated in this activity.

Students interaction with their peer members and the class during the activity was good and effective.

- ❖ ***Challenges faced in implementation:***

Some students are hesitated to share their knowledge in the class. And they are motivated to come out of their fear.

References:

1. <https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html>
2. <https://www.teachingexpertise.com/classroom-ideas/think-pair-share-activity/>
3. <https://www.kent.edu/ctl/think-pair-share>